

A Concise Derivation of Principal Component Analysis

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Goal

Given data points

$$x_i \in \mathbb{R}^p, \quad i = 1, \dots, n,$$

PCA aims to construct a lower-dimensional representation

$$y_i = U_k^T(x_i - m), \quad y_i \in \mathbb{R}^k, \quad U_k \in \mathbb{R}^{p \times k}, \quad k < p.$$

Here, m is the mean of the data, and the columns of U_k define the k -dimensional subspace onto which the data is projected.

The goal is to choose U_k such that the projection preserves as much information as possible.

In PCA, information is measured by variance. Therefore, PCA chooses the projection directions so that the projected data has maximal variance.

Step 1: Center the Data

The mean of the data is

$$m = \frac{1}{n} \sum_{i=1}^n x_i.$$

Define the centered data points

$$z_i = x_i - m.$$

Centering is necessary because PCA should describe how the data varies around its mean, rather than where the whole dataset is located in space.

Thus, the PCA projection becomes

$$y_i = U_k^T z_i.$$

Conclusion

PCA starts from the goal

$$y_i = U_k^T(x_i - m),$$

where y_i is a lower-dimensional representation of x_i .

To preserve as much information as possible, PCA maximizes the variance of the projected data. This leads to the constrained optimization problem

$$\max_v v^T S v \quad \text{subject to} \quad v^T v = 1.$$

Using Lagrange multipliers, this becomes the eigenvalue problem

$$Sv = \lambda v.$$

Therefore, PCA chooses the eigenvectors of the scatter matrix with the largest eigenvalues.

These directions preserve the largest amount of variance and minimize the approximation error.

PCA reduces dimensionality by projecting data onto the directions of maximal variance.